PEP Financial Network OSINT

Employer portfolio brief

Jason Keen | Updated 2026-10-25 | Synthetic portfolio

This portfolio demonstrates a reproducible financial-intelligence workflow using a fictional PEP, companies, banks, transactions and assets. Its central result is an auditable record of what the evidence supports, what it does not establish, and what additional collection is needed. The project is a worked implementation layer; it does not claim universal methodology validity or formal operational intelligence experience.

The analytic question

Which entities, flows, assets and intermediaries can be linked, and what supports ownership, control, access, benefit or mere association? Public-record networks can overstate attribution when different propositions are collapsed into a single relationship. This demo keeps shareholding, family ties, directorship, payments, asset association, succession and title separate.

The workflow

Requirements > entity resolution > ownership and control > transactions > asset linkage > corporate succession and title > confidence and unknowns > collection gaps.

The requirements fix the question, evidence cutoff and synthetic scope. Entity resolution uses stable IDs and registration keys; name similarity never merges records automatically. Dated relationships separate current from historical shareholding. Transaction records retain currencies, purpose and bank context. Asset linkage compares counterparties and timing, while title is assessed separately from legal succession. The validator rejects a chronology-based funding inference if the transfer follows purchase or the payee is not the purchaser. Every claim retains its sources, qualification, competing explanation and next evidence where needed.

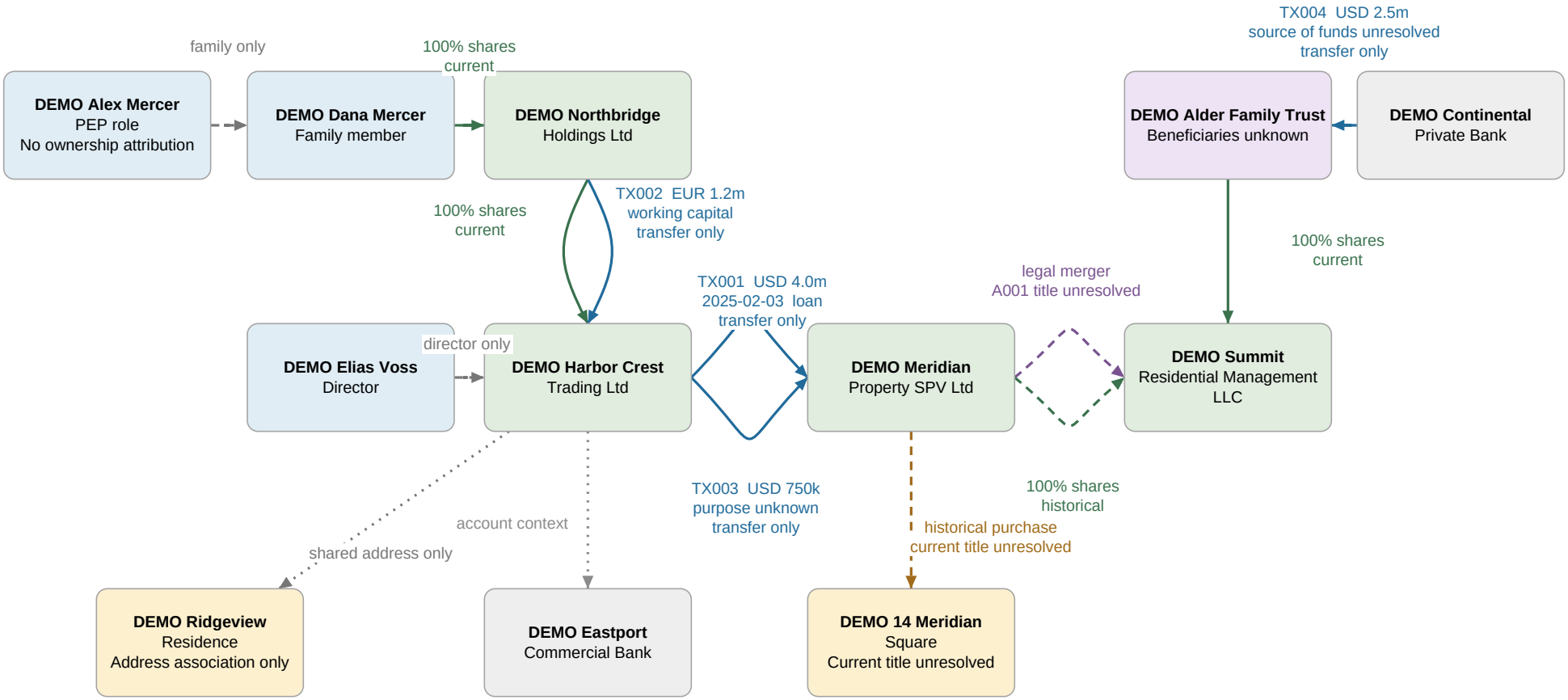
What the demonstration establishes

Dana Mercer is the documented shareholder of Northbridge Holdings. Alex Mercer is a family member, but that relation does not establish his ownership. A USD 4.0 million intercompany loan reaches Meridian Property SPV 17 days before its USD 5.5 million property purchase. The timing is compatible with a contribution to acquisition liquidity; no record earmarks the transfer to the property. A later merger is documented, but current asset-specific title and trust beneficiaries remain unresolved.

Outputs: a typed network graph, nine-claim evidence matrix, source register and dependency table, unresolved collection gaps, validation reports and reproducible inputs.

Synthetic financial network

All names and records depict fictional scenario roles | Evidence cutoff 2026-02-01



Green: shares | Blue: transfers | Purple: merger | Gold: historical purchase

Gray: association or role | Dashed green: historical shares | Dotted gray: address or account context

Read each edge by its type. Family and shared address do not establish ownership. Transfers do not prove property funding. Historical purchase does not establish current title. Merger succession does not identify an asset-specific title transfer.

The graph retains all 12 nodes and 14 typed edges. Rejected ownership/funding propositions and unresolved beneficiary/title claims remain in the evidence matrix; they are not attribution edges. The layout is adapted from the validated demo without changing its nodes or relationships.

Evidence discipline and reproducibility

Confidence and open questions

The claims ledger distinguishes Confirmed, Strongly supported, Probable / analytic judgment, Alleged / unverified, Unresolved and Rejected. Confidence is qualitative and specific to the proposition, not a legal probability. Evidence maturity is separate: Established, Corroborated, Analytic or Open dependency; Experimental is a reserved label with no scoring implementation.

S005 summarizes S004, so the two count as one documented source family rather than independent corroboration. CL007 needs an asset-specific merger transfer schedule and post-merger title extract. CL009 needs a trust deed or verified beneficiary record. TX004 needs originating-account records to resolve the economic source of funds. Rejected means unsupported on this bounded record, not disproven for all time.

Executed validation

Check	Recorded result
Automated tests	31 passed, including all 14 frozen controls
Clean synthetic inputs	0 errors; 1 expected derivative-context warning
Intentionally broken inputs	24 errors; validator exits 1; graph build refused
Generated artifacts	12 nodes; 14 edges; 9 claims; GraphML round-trip passed
Repeated local builds	Input hashes recorded; repeated output hashes match

Technical stack and limitations

Python, NetworkX, CSV and GraphML support a reproducible offline demo, with pinned dependencies, an input SHA-256 manifest and automated GitHub checks. The validator checks IDs, temporal ownership, currency aggregation, funding versus valuation, lineage, attribution and visible dependencies. The CI workflow reruns these controls. A hostile self-review corrected misleading labels and evidence-link weaknesses; it was not an independent audit.

All records are synthetic. Validation cannot authenticate documents, discover undisclosed copying, verify foreign law or establish real-world beneficial ownership. No experimental weights or legal scores are used. A purchase price is not current market value; offshore location and PEP status do not imply misconduct.

Skills demonstrated

Requirements framing, OSINT evidence qualification, due diligence, AML and financial-crime reasoning, entity resolution, transaction analysis, network analysis, uncertainty management and reproducible research. The portfolio shows how explicit evidence boundaries support a reviewable intelligence product and focused next collection.

Implementation reference: jasonrkeen/pep-financial-network-osint (private; access controlled). Execution evidence: outputs/test_results.txt, verification_summary.json and validation reports; interpretation: evidence_matrix.csv, unresolved_gaps.csv and docs/red_team_review.md. MIT License, copyright 2026 Jason Keen. Public visibility remains blocked until the owner gives final approval after release review.